Central Real-time Monitoring System for Premature Baby Incubator

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***Abstract*—It is of great importance to provide a neutral temperature environment that is conducive to the growth and development of the premature babies, who have a small gestational age and low weight and therefore are prone to have various growth and development problems. This paper designs a central real-time monitoring system for premature baby incubator by taking environmental temperature control as the main entry point. The incubator collects environment data by using temperature and humidity sensors and web cameras and displays the data of multiple incubators centrally in the central monitoring interface through networking technology so as to realize real-time and centralized monitoring of the living environment and physical state of multiple premature babies. Therefore, the incubator can effectively reduce the probability of medical accidents and safeguard the healthy growth of premature babies. In consideration of the significant influence of environmental noise on the sleep of premature babies, the monitoring and suppression of noise from incubator will be carried out in further studies.**

Keywords—baby incubator, real-time monitoring system, microcontroller, sensors, medical electronic

I. INTRODUCTION

It is better to put the newborn that have immature physiological functions and incomplete temperature regulation functions in the "neutral temperature" environment, especially for premature babies and stunted newborns. "Neutral temperature" refers to the optimal ambient temperature for maintaining normal body temperature and skin temperature, at which the body uses the least amount of oxygen, loses the least amount of heat through evaporation, and has the lowest metabolism. Premature baby refers to the infants whose gestational age is less than 37 weeks. Due to their small gestational age and low weight at birth, they are prone to have various growth and development problems, and even serious diseases or disabilities.

With scientific methods, the baby incubator creates a comfortable environment with clean air, suitable temperature and humidity for newborns, which can avoid newborn infections, enhance body resistance, and ensure their better development and growth. Baby incubator can provide a good environment with suitable air, temperature and humidity for premature babies and low birth weight infants. It can make the temperature inside the incubator close to the value set by the user automatically, making it especially suitable for premature babies who lack the self-regulation ability of body temperature. Therefore, baby incubator has become a necessary medical device in neonatology department and related departments, and the design of baby incubator has also become the focus of scholars. Virginia Hall designed a solar incubator [1], which uses solar panels to supply power to the incubator, improving its durability and stability. Lichuan Liu [2] designed an active noise control system for baby incubator to reduce the high level of noise within the incubator and provide a safe and quiet environment for newborns in the incubator. Arif Widianto [3] proposed a remote weight monitoring system for premature baby incubator, which could send the weight data monitored by the system to medical experts, so as to evaluate the health status of premature babies. M. Shaib [4] developed a portable incubator for premature babies, which is easy to carry and cheap, aiming to provide security for premature babies in rural areas of low-income countries. Although these designs are very innovative, yet no specific research has been done on the parameter of temperature in the box, and no corresponding preventive measures have been taken against temperature out-of-control in the box.

Adverse events of baby incubator and severe medical accidents caused by newborn burning ascribed to temperature out-of-control of in-service incubator have happened to medical institutions in recent years. Temperature out-of-control of baby incubator has become one of the main nursing injuries to newborns, making it imperative to strengthen the safety monitoring of the incubator temperature. To solve this problem, this system takes the temperature as the main monitoring object and focuses on the real-time monitoring of the temperature of infant and incubator, aiming to minimize the possibility of temperature out-of-control. Khusnul Khotimah [5] developed a remote system that is mounted on a mobile device for the purpose of controlling and monitoring the internal temperature of the compartment baby incubator. Jaroonrut Prinyakupt [6] designed a prototype to verify the temperature and humidity inside the baby incubator, thus improving the sensor accuracy. Muslim Ali [7] proposed a control system, which can monitor the temperature and humidity parameters in the incubator and provide simple monitoring services for users. All of these designs conduct targeted study to the incubator temperature, which, however, adjust the baby incubator. It actually means developing a new product. As a medical device, baby incubator needs a lot of trials that last for several years before putting into clinical use so the existing urgent problems cannot be solved. The monitoring circuit proposed in this paper is independent of the baby incubator and does not affect the normal operation of the baby incubator. The accuracy of its monitoring data will not

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be affected by the state of the baby incubator, and can be compared with the temperature and humidity data displayed in the incubator to ensure the accuracy of monitoring recorded data. This system also integrates the remote video monitoring module [8] and [9]. Through Ethernet and public network, video data can be sent to the monitoring equipment of the hospital and the devices of the premature babies' parents remotely. On the one hand, it helps doctors analyze the physical conditions of premature babies. On the other hand, parents can see their children remotely in real time, reducing the numbers of visit to the ICU.

The rest of this paper has the following structure. Part 2 introduces the hardware circuit of the central monitoring system for premature babies. Part 3 expounds the software design process of the system. Part 4 shows the implementation of the monitoring system for premature babies. Part 5 makes a summary of the design, and a prospect for the next step.

II. Hardware Design

The system consists of upper computer and lower computer. To be specific, the upper computer includes screen display, central station server and monitoring interface designed based on Kingview platform. The lower computer is mainly designed for temperature acquisition circuit, including temperature and humidity sensor, temperature and humidity probe, network camera, etc.

After starting the central monitoring system of premature babies, each data acquisition nodes begins to monitor their premature babies in greenhouse in a real-time manner. Multi-channel real-time temperature and humidity data, body temperature data and video camera surveillance data will also be shown on the display screen of customer and the monitoring room. At the same time, the video data will be recorded in computer hard disk (storage).

If it is necessary to carry out key monitoring on a newborn, the multi-channel real-time monitoring screen can be switched to the one that requires key monitoring, so that only the real-time data and images of the child can be displayed on the large screen. When the temperature and humidity in the incubator or the body temperature exceeds the upper and lower limit of the set temperature and humidity, the sound and light alarm will be triggered to remind the medical staff to deal with it in time. The block diagram of the real-time central monitoring system for premature babies is shown in the diagram below:

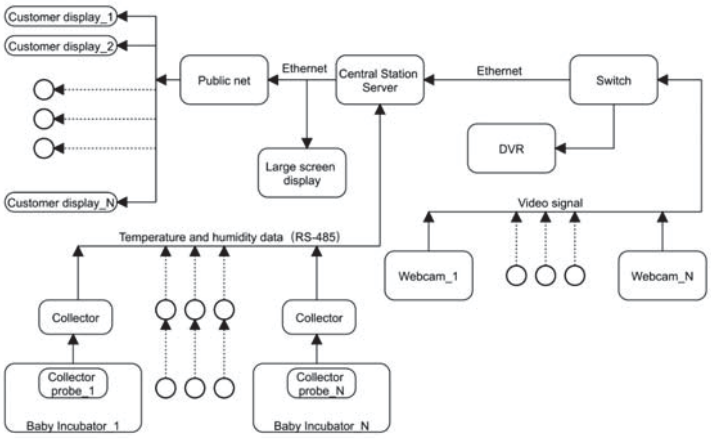


Fig. 1. Real-time central monitoring system

A. Main Control Circuit

In this design, SCM is the core part of the whole hardware system. There are a total of five temperature sensors in each baby incubator, of which two baby thermometers are placed at different positions of the baby's body, and three other box temperature thermometers at different positions in the box. When the system starts, the five temperature acquisition units start to work and save the collected data to the data buffer, and then the SCM sends it to the upper computer. At the same time, the reset circuit, clock circuit and downloading interface are designed. The circuit is shown in the diagram:

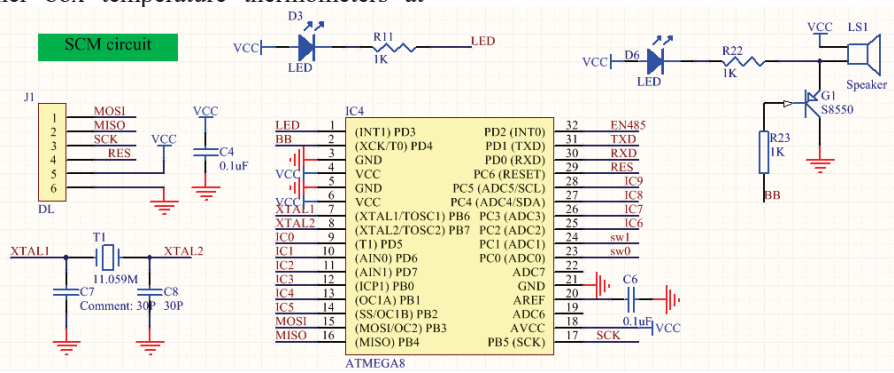


Fig. 2. SCM circuit

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B. Sensor Acquisition Circuit

The temperature acquisition circuit plays a key role in the system. The accuracy of temperature acquisition is related to the infant safety, and thus influences the safety of the infant's development environment. In this design, the digital temperature sensor DS1624 with a storage function is used as the box temperature acquisition chip, the WTics506 chip is used as the baby temperature acquisition chip, and the STH11 is used to collect the humidity information.

The DS1624 chip, which is suitable for low-power dissipation systems and compatible with IIC bus, refers to a digital temperature sensor that integrates measurement and storage. Also, featured by high resolution and stable reading, it has a relatively simple interface with SCM. Therefore, this chip is used as the box temperature and body temperature acquisition unit in this design. Tsic506 is a high-resolution, high-precision temperature measurement chip produced by IST, whose internal part integrates a signal converter for analog or digital signal output. The accuracy is also different in different measurement ranges. It is suitable for the temperature acquisition part in this design. SHT11 series single chip sensor is a temperature and humidity composite sensor containing calibrated digital signal output. This product is superior such as its excellent quality, superfast response, strong anti-interference ability and high cost performance. Therefore, this chip is used as humidity acquisition unit in this design.



Fig. 3. Sensors

This design consists of three box temperature acquisition chips, two body temperature acquisition chips and one humidity acquisition chip. The five temperature acquisition chips will transmit the collected temperature to the upper computer via the communication circuit, and the humidity value will be directly displayed in the incubator, instead of getting involved in transmission. The connection of sensor acquisition circuit is shown in the diagram.

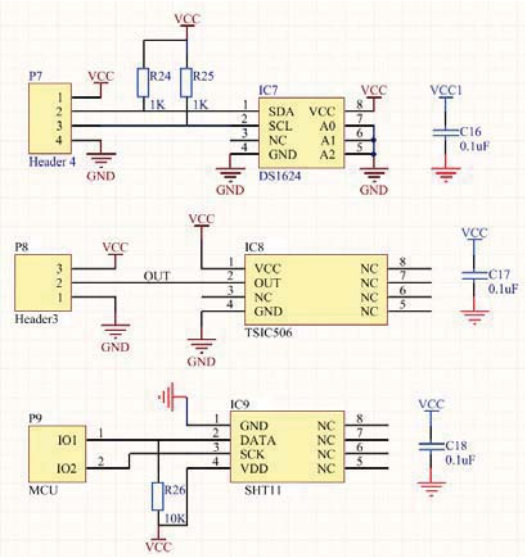


Fig. 4. Temperature acquisition circuits

C. Power Supply Circuit

The power supply circuit provides a stable voltage for the entire system, which is also a necessary condition for the system's normal operation. The performance index of the power supply circuit is directly related to the safety and reliability of the device. The power supply circuit is designed mainly on the basis of considering the input and output voltage limits, the power requirements of the system W, and power dissipation. Finally, 12V DC is used to provide power for the system. The button capacitance and lead capacitance are used as filter capacitance in the design. LM2594 voltage regulator chip is used as the main control chip. The circuit is shown in Fig.5:

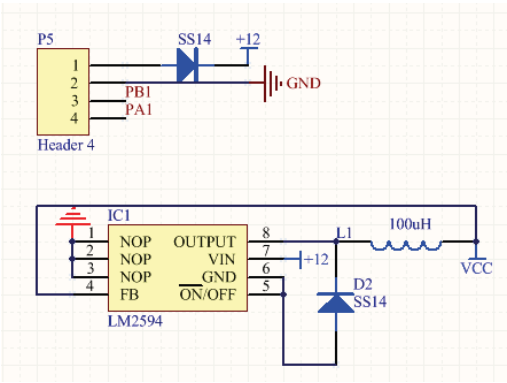


Fig. 5. Power Supply Circuit

D. Communication Circuit

In this design, the communication circuit connects the upper computer and the lower computer in the temperature acquisition part, so that the upper computer can obtain the temperature and humidity value of the lower computer and express it on the display screen. Since this design is aimed at a hospital and there are more than one baby incubators, the hardware interface in the communication circuit uses RS-485 bus technology, and MAX485 is used as the bus driver chip. The communication protocol is realized via the international standard communication protocol ModBus. In this way, 32 devices can be connected on a single bus, which facilitates cost reduction dramatically.

MAX485 chip is used as the driver chip here, which takes half-duplex communication. The same physical channel is shared by sending and receiving. When working, MAX485 chip converts TTL level to RS485 bus level, and uses RXD on ATmega8 to connect TXD port to upper computer through MAX485. As the PC only supports RS-232 interface only, RS-232 to RS485 bidirectional conversion interface is added in the design to realize communication. The circuit diagram is shown as follows:

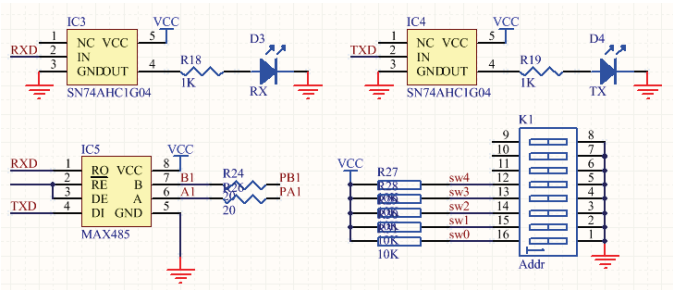


Fig. 6. Communication Circuit

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E. PCB Bottom Plate

In this design, the used components are firstly encapsulated according to the actual standard size, and then the components are laid out and automatically wired. Finally, the designed circuit board is adjusted manually according to the size and shape for the purpose of manufacturing circuit board. The final generated PCB board is shown in the diagram below.

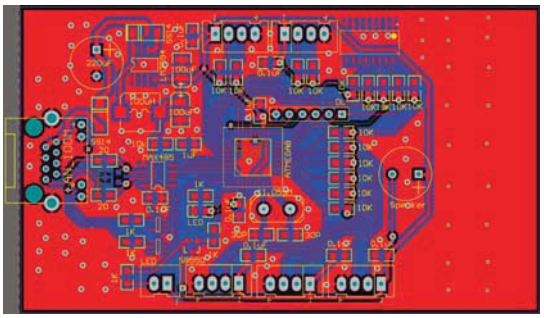


Fig. 7. PCB Bottom Plate

III. Software Design

The software of this design also includes the upper computer and the lower computer. To be specific, the upper computer is fitted with a systematic monitoring interface and the lower computer has a temperature detection and video acquisition program. When the system starts running, the user logs in and configures the relevant parameters first. The setting here means setting the address of each baby incubator. Next, the sensor data and video data in each incubator are stored in the data buffer area. While the system is running, the upper computer sends Modbus data to request communication. After the device address correctness is verified by the SCM, the corresponding operation is performed according to Modbus data frame content, and the corresponding data is sent to the upper computer and displayed in the configuration interface. The alarm program will be started if the temperature data exceeds the limit during the display process. If the temperature is normal, it will return and wait for the next communication request sent by the upper computer. The main program flow chart of the upper computer in the central station is shown in the diagram:

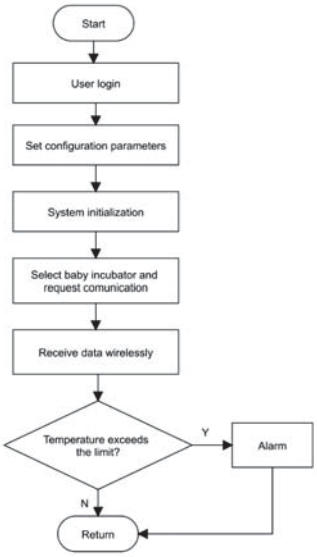


Fig. 8. Flowchart of the upper computer

The temperature acquisition part of the lower computer is a kind of software designed developed based on C language. The flow chart is shown in the diagram. It is mainly aimed to realize the following functions. First, it stores the temperature values collected by the five temperature sensors in the data buffer area. Second, it receives the query instruction from the master device, that is, it sends Modbus data frame to the slave device, completes the analytical verification, and obtains the address W of the slave device and the function code to be executed according to the Modbus data frame contents. Third, it makes corresponding response according to the inquiry instruction of the master device. Fourth, it sends a reply to the upper computer.

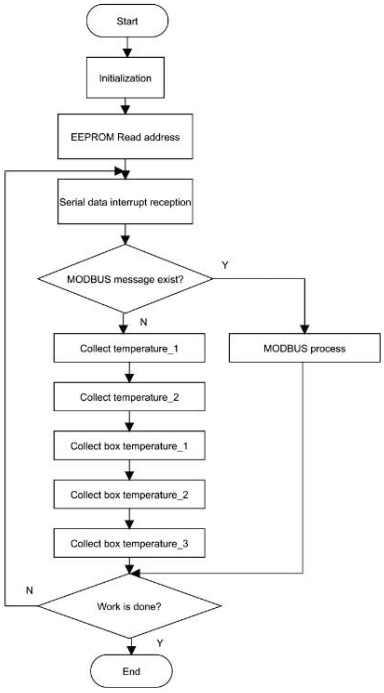


Fig. 9. Flowchart of the tempareture sensor

The flow chart of video acquisition on the lower computer is shown in the diagram:

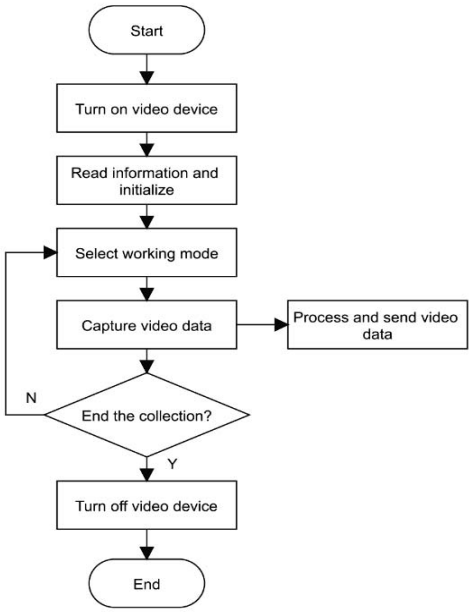


Fig. 10. Flowchart of video acquisition

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IV. System Implement

Baby incubator part of the entire system is assembled as shown in the figure below:



Fig. 11. Premature Baby Incubator

Real-time monitoring demonstration pictures:

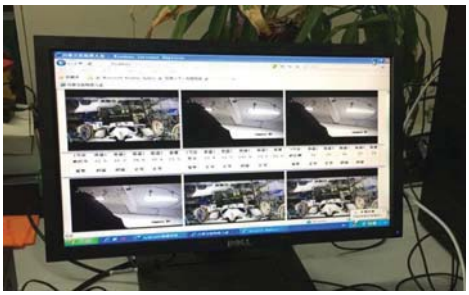


Fig. 12. Real-time monitoring interface(a)

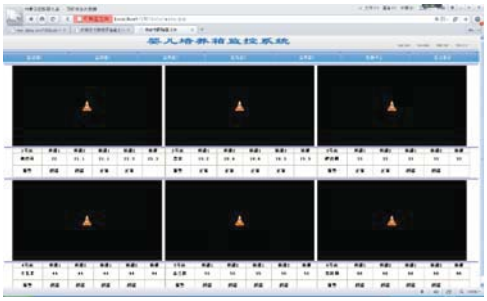


Fig. 13. Real-time monitoring interface(b)

V. Results and Discussion

By using both software and hardware, this paper designs a set of real-time central monitoring system for premature babies that can collect infants' real-time temperature, humidity and video information and realize following functions:

(1) Make the monitoring temperature data continuous, accurate and real-time, and increase device safety. The monitoring circuit of the central real-time monitoring system for premature babies, which is independent of the baby incubator, does not affect the incubator's normal operation. The accuracy of the monitoring data will not be affected by the state of the baby incubator, and can be compared with the temperature and humidity data displayed in the incubator to ensure the monitoring record data accuracy. In practice, there is a difference between the incubator temperature and the baby's skin temperature, and doctors generally concern the most body temperature (skin temperature). The real-time central monitoring system for premature babies provides real-time data of infant skin temperature and box temperature, which ensures the continuity and accuracy of temperature recorded data, and provides clinical guidance for doctors' diagnosis and treatment.

(2) Ensure the normal humidity in the incubator to provide a suitable growth environment for newborns.

(3) Monitor the physical state of the newborn in the baby incubator in real time remotely, reducing the possible risk caused by the abnormal physical state of the newborn. The parents of the premature babies can also see the real-time picture of the child remotely, reducing the number of visits to the ICU and improving ICU's environmental quality.

(4) Improve the working efficiency of medical staff and reduce their error rate. The central monitoring system for premature babies can simultaneously display the newborn's skin temperature, incubator temperature and humidity, and personal information of the newborn in each baby incubator on the central display screen of the nurse station or the central monitoring room. All kinds of graphs and tables needed by medical staff can be automatically analyzed and generated, which greatly reduces the labor intensity of medical staff, improves the nursing efficiency and reduces their error rate.

However, this design also has some aspects to be improved. For example, the baby incubator may cause excessive noise due to device aging. Under normal circumstances, the noise level in the incubator should not be higher than 55dB. A too high noise affects the normal development of premature baby's auditory sense. Therefore, the design will be further improved by setting noise detection and suppression device. In this design, the video information receiving terminal of the newborns' parents is limited to PC and will be accessed to mobile terminal through further improvement, so that the parents can login and access the receiving terminal in the WeChat Official Account.

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